AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation log

Single living document. Superseded claims are struck through, not deleted.

Last updated: 2026-07-13

Abstract

Where we are (2026-07-14, after the fixes). Two real, independent bugs explain the catastrophes of the past week; both are now fixed and under test (movcav16):

Bug I — PISM's ocean forcing was surface water. The `-ocean th` route averaged FESOM level indices 1–20 = **0–410 m including the summer surface**: PISM received ice-base water of up to +6.5°**C** even from a healthy ocean, melted accordingly (∼30 m/yr), and shed **24% of its floating area per decade** — so every mesh-change leg had to absorb a catastrophic geometry increment. FESOM's *own* cavity melt in the same healthy runs is sane: **2.2 m/yr mean, 0.72 median**. Fix: PISM now consumes FESOM's own melt directly (`-ocean given`, the DIRECT route; esm_tools `93a207ee`).

Bug II — the offline feom exchange weights were in the wrong node ordering. FESOM exchanges coupling fields in *partition (my_list/rank) order*; the offline weight engine (2026-06-17) built them in *nod2d.out order*. Result: every A096↔feom exchange in every submesh leg was **geographically tile-translocated** — the atmosphere was coupled to a jigsaw ocean (Fig. 1): OIFS saw Weddell ice fraction **0.04** while FESOM had **0.95**; global sea-ice death followed within months. Fix: permute to runtime ordering + a mandatory weight validator that aborts the leg on ordering mismatch (`c21fe7c2`).

Four earlier root-cause claims are falsified and struck through below. Sections whose measurements were made on jigsaw-coupled runs are marked **TAINTED** and will be re-measured under movcav16.

Contents

1 The two real bugs (2026-07-14)

1.1 Bug I: PISM was fed surface water as ice-base forcing

`coupling_ocean2pism.functions` built PISM's `-ocean th` forcing as a vertical mean over FESOM level *indices* 1–20 (`-sellevidx,1/20`) = **0–410 m, surface included**; the correct `-sellevel,150/600` sat in dead code behind `if [ 0 -eq 1 ]`. Measured consequence in the *healthy* chain (movcav8): `theta_ocean` fed to PISM had mean −0.43°C and **max +6.53°C** (realistic ice-base: ≈ −1.9). PISM's three-equation scheme melted from that at ∼30 m/yr and collapsed the shelves: −24% floating area per decade, 1399 nodes losing their ice *entirely* per 10-yr leg, Ross front −440 km (Fig. **??**). Meanwhile **FESOM's own melt** in those same healthy runs: **2.2 m/yr mean, 0.72 median, deep-cavity 2.6** — sane.

Fix (`93a207ee`): the DIRECT route. PISM consumes FESOM's own cavity fluxes via `-ocean given`: `fw`→`shelfbmassflux` (×1000; sign preserved — FESOM melt is $fw < 0$ and PISM's `shelfbmassflux` is "positive if ice is freezing on"), `sst`(level 1 = ice base under the virtual-column convention)→`shelfbtemp`. One model owns the ice–ocean interface. The `-ocean given` machinery existed all along; it silently fell back to `th` because its input is a FESOM1.x-era file (`EXP.year.pism.nc`) that FESOM2 never writes — `fesom2ice` now synthesizes it from the standard monthly streams. The 150–600 m band is also enabled as a safety for anyone still on `th`.

1.2 Bug II: the exchange weights scrambled the coupled ocean

FESOM registers its OASIS partition rank-contiguously (Apple partition, `cpl_driver.F90`): the exchange field is the concatenation of each rank's owned nodes in `my_list` order — **not** `nod2d.out` order (median displacement between the two: 58° of latitude). The offline weight engine introduced on 2026-06-17 built the feom geometry in nod2d order. The weights were validated structurally (dims, layout) but never *applied to a field* — and at runtime every exchange was tile-translocated. The canonical pool weights are correct because they were *harvested from a real coupled run* (hence the per-nproc pool directories).

Evidence (all on movcav15 itself): OIFS ci = 0.04 vs FESOM a_ice = 0.95 in month 1; 1920 tropical cells < 5°C and 746 polar cells > 15°C in OIFS's January SST (0 and 0 in movcav4); offline application of the exact staged weights to the exact FESOM ice field reproduces OIFS's received 0.04; global sea-ice death with **no winter regrowth** in both hemispheres; cavity +1.25 K, melt 28 m/yr *regardless of initial state* — the heat arrived through the scrambled exchange every timestep. Every submesh leg since June carried this (masked there by the day-4 crashes); the healthy chunk-1 runs (movcav4/7/8) used pool weights.

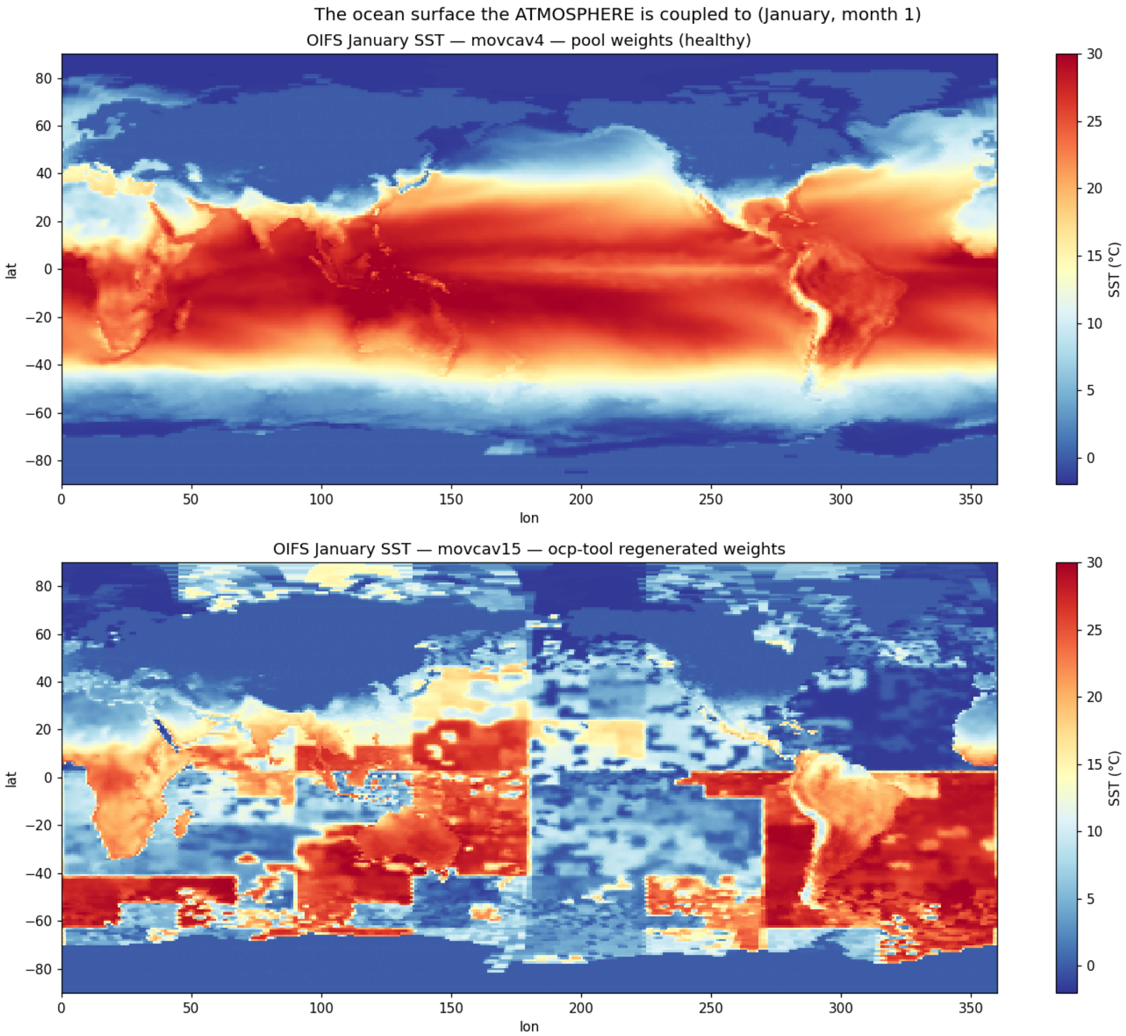


Figure 1: **The picture that settles it.** January (month 1) sea-surface temperature as OIFS holds it. Top: movcav4, pool weights — textbook. Bottom: movcav15, offline-engine weights — the ocean is a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a straight seam at lon 180 where the mesh node ordering breaks.

Fix (`c21fe7c2`, option B1 — in the coupling layer, ocp-tool untouched): `permute_feom_to_runtime.py` reorders `feom.*` in grids/masks/areas to runtime order (from the submesh's own `dist_<nproc>/rpart.out`, verified identical to the `my_list` concatenation FESOM reads); `validate_feom_weights.py` then applies every weight file to the runtime-ordered node latitudes and **aborts the leg** (couple_fail) if the median error exceeds 1° (nod2d-ordered weights fail at 42–58°; correct ones pass at $\leq 0.025°$). The validator would have caught the bug the day the engine was introduced. Round-trip tested on compute; the pooled chunk-1 artifacts are fixed in place (backups kept).

1.3 How the two bugs interlocked

Bug I collapsed PISM even in healthy runs $\Rightarrow$ monstrous per-leg geometry increments. Bug II poisoned the runs used to diagnose Bug I: measured on jigsaw-coupled movcav15, "FESOM's own melt" appeared to be 28 m/yr, which falsely disqualified the DIRECT fix and sent the investigation into the cavity-water/SPP dead end. **Measurements from a corrupted run are worse than no measurements.**

2 Blocker I — OIFS step-0 coastline crash (RESOLVED)

A floating-point NaN struck exactly the coastline cells whose land–sea type flipped. `ocp-tool` half-converted a flipped cell (only mask + soil type), OIFS re-ingested the inconsistency from the re-generated `ICMGG` and NaN'd in moist physics. Fixed by a masked nearest-neighbour rebuild of each flipped cell's full column, plus the `ICMGG` re-ingest (`PUTALLFLDS_M` in `reresf_part2.F90`). A recurring `voskin` floating-overflow variant shows up state-dependently and was traced to the native-pipeline `ICMGG` coastal lsm.

3 Blocker II — FESOM cavity-margin blowup

3.1 Two root causes we got wrong

FALSIFIED #1 — "a geostrophically-unbalanced remapped state". ~~The mesh change plants fresh cavity fronts against salty open water but zeroes the balancing current; initialise the changed-cell velocity in thermal-wind balance instead of zero.~~ That fix was implemented (per-level $\mathbf{u}_g = (1/f\rho_0)\,\hat{z}\times\nabla p$, using FESOM's own cavity pressure convention) and **failed its own offline gate**: the discrete ∇p at grid-scale fronts is *noisier* than the nearest-neighbour fill. It ships opt-in only (`REMAP_GEOSTROPHIC=1`), off by default.

FALSIFIED #2 — "the sub-ice free-surface BC violation". ~~FESOM keeps $\eta \equiv 0$ under ice, yet the remap gave the 987 newly-sub-ice nodes their old open-ocean ssh (≈ -1.6 m); fixing that will cure the blowup.~~ A **real bug**, and fixed (esm_tools `b17fb5c8`) — but **not the cure**: the fixed run still died on day 4.4.

3.2 Every hypothesis, and the experiment that killed it

Hypothesis	**Experiment**	**Verdict**
Atmosphere / OASIS / coupling	ocean-only standalone FESOM reproduces the crash exactly (day 4, CFLz 9.24)	**ruled out**
CORE2 forcing shock (standalone artefact)	full-mesh control, *same* forcing + native restart: stable 21 d, worst CFLz **2.61**	**ruled out**
Remap corrupting the state	remapped vs. source over the 204 970 unchanged nodes: max $\lvert\Delta\rvert = \mathbf{0.000}$ for `temp`, `salt`, `ssh`, `hnode`	**ruled out** (bit-exact)
The restart *fill values* at changed nodes	five independent fill strategies (donor, coherent-column, shelf-base hold, NN velocity, full balance hygiene)	**ruled out** (all die ≈day 4)
Pathological thin water columns	submesh min wet column 30 m vs. mother's 25 m; zero 1–2-layer columns	**ruled out** (submesh is *cleaner*)
Basal melt flux shock	`cavity_flux_ramp_days` implemented; melt effectively **OFF** still dies (mstep 328)	**ruled out**
Timestep / unresolved w	1200 s: CFLz kill day 4.0. 60 s: **0** CFLz warnings, dies day 4.4 via η_n. 120 s: dies day 8.9	**ruled out**
Sub-ice free-surface BC	real bug, fixed; run still dies day 4.4	**not the cure**
Cavity geometry itself	**cold start on the same PISM submesh @1200 s: STABLE, day 13+, worst CFLz 3.44**	**ruled out**

3.3 The decisive control: a cold start on PISM's cavity is stable

Same submesh, same forcing, same production timestep — only the initial state differs:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up CORE3 restart	stable 21 d, worst CFLz **2.61**
PISM cavity, cold	**1200 s**	**cold (climatology, at rest)**	**stable 13+ d, worst CFLz 3.44**
PISM cavity, warm	1200 s	our remapped restart	**dies day 4.0**, CFLz 9.24
PISM cavity, warm	120 s	our remapped restart	dies day 8.9 (η_n)
PISM cavity, warm	60 s	our remapped restart	dies day 4.4 (η_n, 0 CFLz)

The ocean has no difficulty living in PISM's cavity at the production timestep. Whatever is wrong is in *our restart*, not in the mesh, the physics, or the coupling.

3.4 What is actually wrong: the dynamics seam

The remap produces an internally inconsistent state: on the **unchanged nodes (98%)** it reproduces the evolved restart **bit-exactly** (a fully developed, balanced dynamic state: $\mathbf{u}$, η, AB history); on the **changed nodes** it substitutes *patched* values. The two regions are mutually incompatible, and the **seam between them** is the instability. Zeroing the dynamics globally removes the crash (the ocean re-derives its own balanced flow from rest) — which is exactly why all five fill strategies failed: each varied the *patch* while leaving the *seam*. Zero-dynamics is not a production fix: discarding the ocean's circulation every leg is unacceptable.

4 The increment test (2026-07-13) — TAINTED, to be re-run

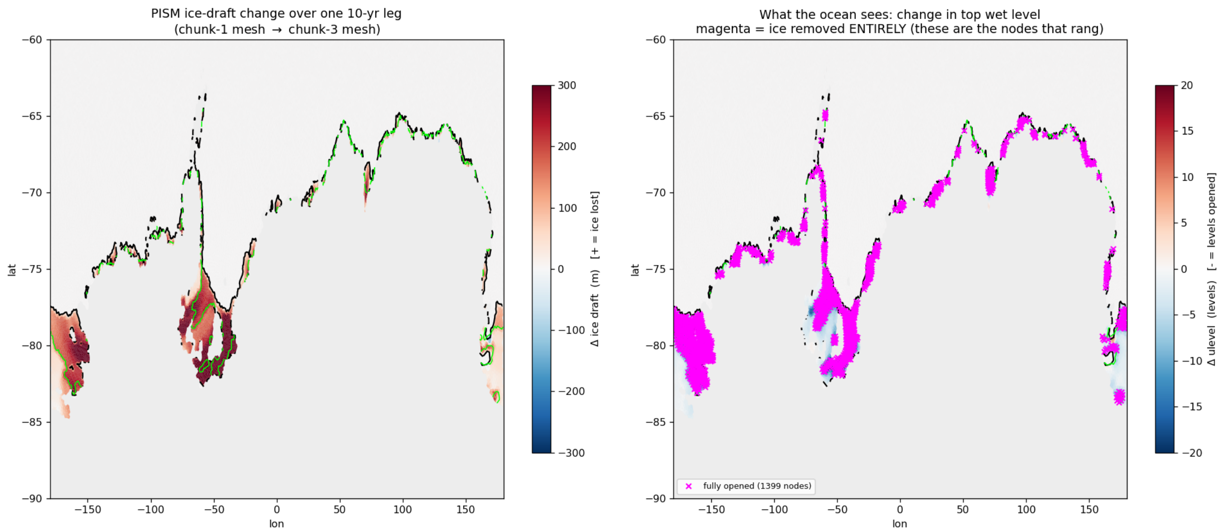
TAINTED: everything in this section was measured on movcav12, whose chunk-1 climate was jigsaw-coupled (Bug II) and whose PISM leg was additionally driven by Bug I forcing. The qualitative machinery findings (the remap carried the increment; the failure mode was 17 fully-opened columns) may survive, but no number here should be trusted until the test is repeated under movcav16.

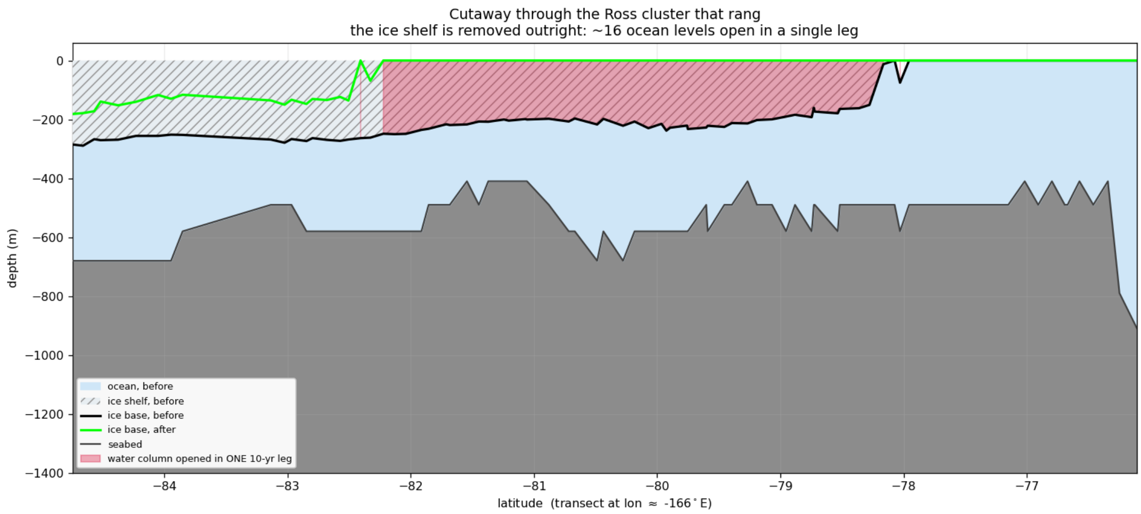
Cold-starting the ocean on PISM's cavity sidesteps the one-off CORE3(observed)→PISM swap ($\max|\Delta\text{draft}| = 2711$ m) entirely. That was done: chunk 1 ran a clean model year, PISM ran its 10 years, and the remap was finally asked to carry *a year of spun-up ocean dynamics across a real PISM increment*. It crashed on day 15.7.

	one-off swap (CORE3→PISM)	**real increment**
CFLz over time	4.46 → 6.43 → 7.25 → 9.24	5.26 → **5.78** → 5.40 → **4.15**
	monotonic runaway	**peaks, then decays**
died at	day ∼4	day 15.7
η_n NaN	yes	**none** — all 209 172 nodes finite
$\|\eta\| > 3$ nodes	3611, basin-wide	**17**, one cluster, **all changed nodes**

All 17 ringing nodes share one signature: `ulev 17 → 1` — **16 ocean levels opening at once**, i.e. the ice shelf above them removed *entirely*. Their η alternates in sign ($+24.4, -17.9, +15.9, +8.3$): a grid-scale **checkerboard**, not a basin mode.

4.1 The finding: PISM is collapsing





The cutaway is the clearest statement of the problem: the ice base goes from $-200\ldots-280$ m to **zero (open ocean)** across the whole span $-82.2°$ to $-78.2°$ latitude. **The Ross ice-shelf front retreats ≈440 km in ONE 10-yr leg.** The plan view shows this is not a Ross anomaly: the 1399 fully-opened nodes (magenta) ring the *entire* Antarctic margin.

changed nodes this leg	3248	(1.55% of mesh)
$\lvert\Delta\text{ulev}\rvert \geq 10$		1582
nodes with the ice removed ENTIRELY		**1399**
~~mean $\lvert\Delta\text{draft}\rvert$~~	~~2.9 m~~	← **meaningless**
max $\lvert\Delta\text{draft}\rvert$		**923 m**
PISM floating cells	24 815 → 20 093	(**−24% in 10 yr**)

FALSIFIED #3 — "the per-leg increment is gentle". ~~Production never has to survive the one-off swap; the per-leg PISM increment is two orders of magnitude smaller (mean |Δdraft| ≈ 22 m) and therefore absorbable.~~ **Wrong, and wrong in a way that mattered** — it motivated the entire "sidestep the swap and the increments will be fine" strategy. The mean is diluted by the 98% of the domain that does not change; the distribution is wildly skewed. PISM is *not* in equilibrium.

4.2 This is not good news

It is tempting to read the numbers optimistically: only 17 of 1399 fully-opened nodes rang; the ocean absorbed ∼99%; the CFLz transient peaked and *decayed*. All true, and it does mean the remap machinery is performing better than a "crashed at day 15" headline suggests.

But the honest reading is negative. The run still crashed. What we have learned is that **our ocean reacts violently to ice-shelf and cavity changes mid-run**: a handful of columns where the ice vanishes outright is enough to kill a leg. A coupled setup in which the ice sheet is *allowed* to change its cavity every 10 years cannot be one leg's worth of extreme nodes away from failure. **Absorbing 99% of the change is not a passing grade when the remaining 1% is fatal.**

5 Falsified: the SPP claim

FALSIFIED #4 — "the salt plume parameterisation was off". ~~spp=.false. means no brine plume, no HSSW, warm shelf — turn spp on.~~ Checked against the numbers instead of the figure: in Xiaojie's CORE3 runs the **spp-OFF cavity is the coldest** (thermal driving +0.177 K vs +0.246 with spp on), and CORE3 standalone keeps a cold cavity *with our exact setting*. The warm cavity in movcav12/15 was Bug II (the jigsaw), not spp. The claim was made from reading a shelf-transect figure without checking whether the signal reached the cavity. Config change reverted.

6 Where to go from here

movcav16 is the clean end-to-end test: fixed weights + warm-tracer init + DIRECT melt coupling. Verification checklist for chunk-1 (each item targets one of this week's failure modes):

1. OIFS January SST map is physical (the jigsaw figure *is* the test);
2. OIFS `ci` ≈ FESOM `a_ice` (was 0.04 vs 0.95);
3. sea ice survives the winter in both hemispheres (was: dead by April, no regrowth);
4. FESOM cavity melt ≈2 m/yr (was 28); cavity thermal driving ≈0.2 K (was 1.25);
5. PISM leg: floating-area change per decade ≪ 24%;
6. then chunk-3: the *real* increment test, finally on clean evidence.

6.1 movcav16 launch log (2026-07-14)

Three attempts were needed to get movcav16 airborne; none of the failures indicts the two fixes.

- **Attempt 1** (job 26245041): healthy to day 120 (worst CFL_z 2.41, EVP active, ∼17 d/min) – killed by a launcher bug of ours (`env_fesom.py` looked up `oce2ice_method` under the setup name instead of `config['fesom']`; fixed in esm_tools `d5a8039d`). The teardown of the crashed launcher cancelled the healthy compute job.
- **Attempt 2** (job 26245420): real model crash. OIFS floating overflow in the convection scheme (`cubasen.F90:453`, rank 29) on 19 January; the watchdog then cancelled the job correctly. FESOM was healthy. The work dir's `rmp_*feom*` files are bit-identical to the corrected pool weights, so the ordering fix *was* in effect. Attempt 1 ran the same inputs past day 120, so the overflow is *not deterministic*: a marginal convection state that reproducibility noise (OpenMP physics scheduling) flips. `BETAMFBC=-1` was already set (not the qmfixer

overflow). Noted for the suspect list should it reproduce: chunk-1 uses the native-pipeline `ICMGGab45INIT_feomdyn`, which hardens 74 Antarctic coastal cells from fractional ocean to full land vs. movcav4's clean CORE3 file (intentional – the cavity coastline; soil fields on those cells are sane glacier-convention values). Forensics: `movcav16_crash1_forensics/`.

- **Attempt 3**: relaunched from the iterative master runscript (`spinup_couplePI_core3.yaml`; launching the model1 sub-runscript directly fails with `KeyError: model1`). `use_venv: False` added to the sub-runscript so non-tty launches cannot hang on the venv prompt.

7 Bugs found and fixed along the way

7.1 In the remap

Unbounded extrapolation below the old seabed (esm_tools `eef4f6db`). The first attempt at the increment leg died at **mstep 1**: the remapped restart contained -45.02°C, a value present *nowhere* in the source (wet range $[-2.32, 41.72]$, dry cells 0.0). The remap *manufactured* it. The column shows the mechanism: $1.22, -0.99, -3.19, -5.39, -8.69, -13.09, -18.6, -25.2, -34.01, \mathbf{-45.02}$ — smooth and accelerating. For columns that get *deeper*, the remap fitted a straight line through the last two old levels and continued it downward with no clamp. Fixed by taking such water from the nearest old node actually wet at that depth. **Never hit before** because every previous remap went mother → submesh (a strict *subset*); ice *retreating* grows the mesh and exercises this path for the first time.

Also: sub-ice `ssh`/`hbar` zeroing (`b17fb5c8`); a **size guard** so the remap refuses a restart that does not match the old mesh, instead of silently writing garbage (`3b44a704`).

7.2 In the coupling plumbing

Making chunk 1 a *submesh* leg pushed a submesh leg through `couple_out` for the first time. Every previous run either ran chunk 1 on the *full mesh* (where these paths are trivially correct) or died early in chunk 3 (before reaching the end of a leg). That exposed:

- **OASIS `GSSPOS` abort at ∼day 300.** Its conservation correction is built from the summed source/target ratio; when the summed target residual is ≈ 0 while the source residual is small-but-nonzero, it explodes. Only the heat-to-ice flux (`A_Q_ice` → `heat_ico`) used it. Fixed: new `gauswgt_gsmart` transform (`3e94d4da`).
- **The namcouple `feom` dimension had NEVER been patched, on any leg.** `fix_namcouple_feom_dim` read `${MAXMESH_DIR_fesom}`, which is assigned *inside* `build_submesh` — a different subjob's shell. Always unset ⇒ guard refused ⇒ silent no-op. Fixed (`3cf29e9e`).
- `fesom2ice` **loaded the static full mesh** while FESOM's output is on the submesh (broadcast error) (`63cfcc91`).
- `previous_submesh` **is read but never created** — anywhere. Every leg from chunk 5 on would have remapped against the wrong mesh (`3b44a704`).
- **A failed `couple_in` did not stop the workflow.** The coupling function is backgrounded, the observe job is not its parent (so it can never see the exit code), and `observe.py` returns an *empty* error list for every non-compute job. The next leg started on an un-remapped restart. Fixed with a `.coupling_failed` sentinel (`3b44a704`).

The pattern: the guards were right, and the plumbing around them defeated them. `remap_oasis_restart` correctly refused a mis-sized restart — and the run proceeded anyway.

8 Assets

- **A 2-minute offline reproducer**: standalone ocean-only FESOM on the PISM-cavity sub-mesh, 14 nodes, crashes in ∼2 min (vs. the 45-node coupled system). Requires the `FESOM_COUPLED=OFF` build at the current SHA.
- **A stable control**: full mesh + native restart + same forcing (21 d, CFLz 2.61).
- **Chunk-1 artifacts pooled** at `input/fesom2/pism_cavity_ini/` (540 MB), so chunk 1 needs no `couple_in` and `esm_runscripts` can be run straight from the login node.
- `cavity_flux_ramp_days` (FESOM `&run_config`, default 0 = off, bit-identical).
- `verify_balance.py`: offline balance/stability gate (min N^2, max $|\nabla \cdot (H\bar{\mathbf{u}})|$, CFL, Ri) — it caught the geostrophic ∇p init being worse than the NN baseline *before* it cost a model run.

9 Method note

Three root causes were asserted with more confidence than the evidence supported, and all three were later falsified by a control that was cheap and should have come *first*:

1. "geostrophic imbalance" — killed by the **cold start** (is it the geometry or our state?);
2. "the sub-ice ssh BC" — killed by simply **running the fix**;
3. "the increment is gentle (≈22 m)" — killed by **plotting the distribution** instead of trusting a mean over a domain that is 98% unchanged.

The third is the sharpest: the 22 m figure was computed correctly and reported honestly, and it was still misleading enough to steer the strategy for a day. **Two figures did in ten minutes what a summary statistic obscured for a day.**